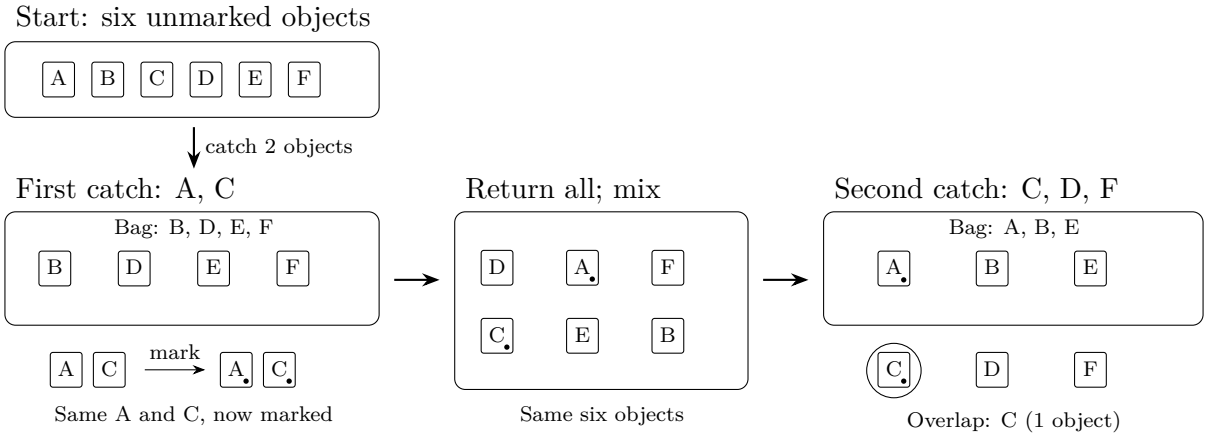
Use identical-size cards or counters with visible marker dots that cannot be felt. Keep the same objects throughout a series; add, remove or exchange none except for temporary catches. For sampling trials, draw without looking; every group of the chosen size should have the same chance, whether marked or unmarked.

Keep each catch out until it is counted. Mark the first catch, return *all* of it, and mix before the second catch. Marks stay on through both catches. An overlap is a marked object in the second catch. Before a new round, return everything and clear the marks. To show a record is possible, you may arrange catches by hand from the same bag; no object appears twice in one catch.



Problem 4: These example rounds all use one unchanged bag, with first catch 6 and second catch 6. Could a bag of 18 objects produce every record? Decide what you would report about the total after seeing all five rounds.

Round	Overlap	Rule's guess
1	1	36
2	2	18
3	3	12
4	2	18
5	0	No matching total

Problem 5: The last round caught no marked objects. The keeper says the bag holds at most 20. Which catch sizes make an overlap of 0 impossible? Explain.

Problem 6: A bag has 24 objects. You may handle 12 objects altogether: first catch size + second catch size = 12. Choose the split with the greatest average overlap over many rounds. Explain why no other split does better.

The average over all equally likely second catches is

$$\text{average overlap} = \frac{\text{first catch size} \times \text{second catch size}}{\text{bag total}}.$$

Problem 7: Would the best split change with a budget of 13 objects? What about 20? Find a rule for any whole-number budget from 2 to 24 and explain why it works.

Problem 8: A keeper says, “Our overlap was small, so our bag must be large.” Some marks might have rubbed off. How could that change the guess? Design a way to test whether lost marks, rather than the total, caused the small overlap.